CODEALPHA INTERNSHIP PROGRAM — BIOINFORMATICS

TASK 2: Multiple Sequence Alignment & Evolutionary Tree Reconstruction

Project Identifier:	CodeAlpha_Task2_MSA_Analysis
Target Biomolecule:	Preproinsulin across Mammals & Teleost Fish
Analytical Software:	Clustal Omega v1.2.4 (EMBL-EBI Framework)
Number of Taxa:	7 Species (Homo sapiens, Primates, Mammals & Outgroup)

1. Introduction & Biological Background

Multiple Sequence Alignment (MSA) is a fundamental computational procedure used to align three or more biological sequences (protein or nucleic acid) of physiological similarity. By aligning preproinsulin amino acid sequences across diverse vertebrate taxa (*Homo sapiens*, *Pan troglodytes*, *Macaca mulatta*, *Bos taurus*, *Sus scrofa*, *Mus musculus*, and *Danio rerio*), we can identify structural conservation, evolutionary drift, and conserved motifs responsible for hormone functionality and receptor interaction.

2. Methodology & Selected Taxa

2.1 Software and Parameters

- **Algorithm / Tool:** Clustal Omega v1.2.4 (EMBL-EBI)
- **Input Dataset:** 7 Vertebrate Preproinsulin Sequences (FASTA format)
- **Color Scheme:** Clustal2 physicochemical property palette
- **Distance Matrix & Tree Algorithm:** Neighbor-Joining / Phylogram calculation

2.2 Species Selected for Evolutionary Comparison

Organism Name	Common Name	Taxonomic Clade
Homo sapiens	Human	Primates (Hominidae)
Pan troglodytes	Chimpanzee	Primates (Hominidae)
Macaca mulatta	Rhesus macaque	Primates (Cercopithecidae)
Sus scrofa	Pig	Artiodactyla (Suidae)
Bos taurus	Bovine / Cattle	Artiodactyla (Bovidae)
Mus musculus	Mouse (Ins2)	Rodentia (Muridae)
Danio rerio	Zebrafish	Teleostei (Cyprinidae) — Outgroup

3. Multiple Sequence Alignment Analysis

3.1 Text & Physicochemical Alignment Output

The alignment output provides detailed insight into structural conservation across the preproinsulin peptide chain. Conservation key annotations are defined as: (*) Single fully conserved residue, (:) Strong conservation of physicochemical properties, and (.) Weak conservation.

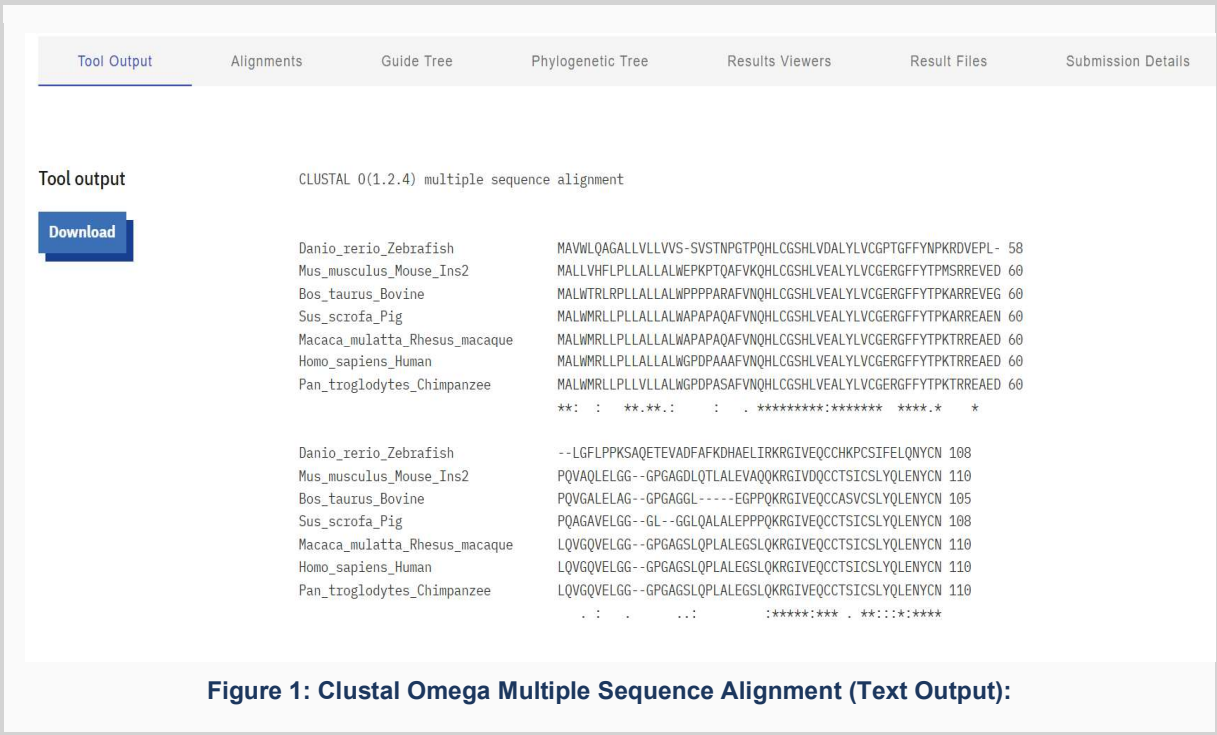


Figure 1: Clustal Omega Multiple Sequence Alignment (Text Output):

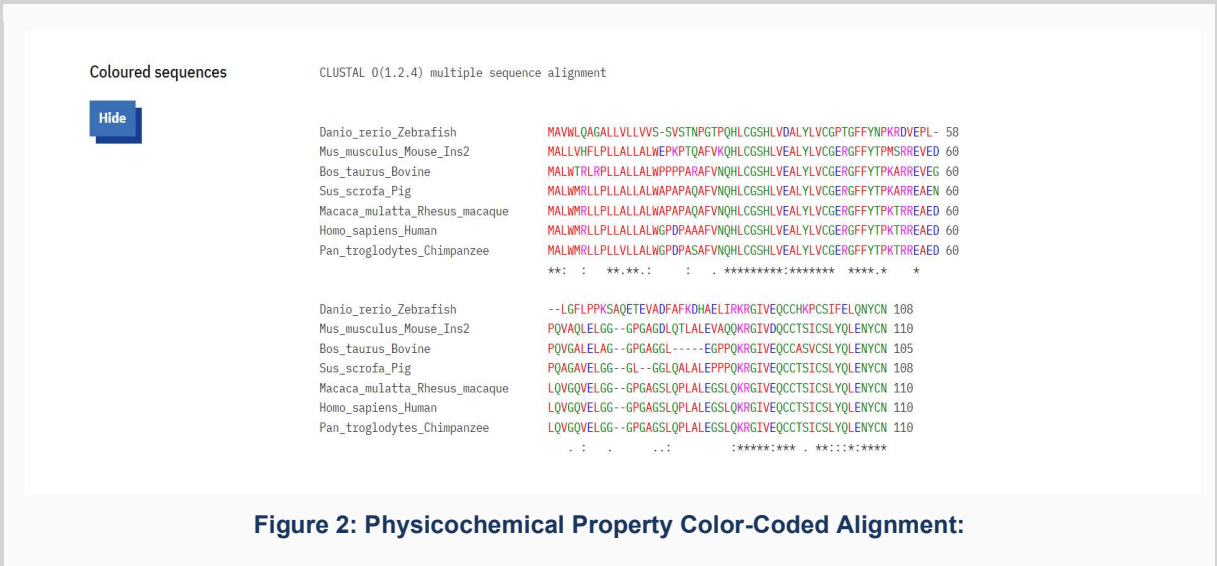


Figure 2: Physicochemical Property Color-Coded Alignment:

 Signal Peptide Region (Residues 1–24): Exhibits variability accommodating hydrophobic substitutions necessary for membrane translocation.

 Insulin B-Chain (Residues 25–54): Displays strict evolutionary conservation across all mammals.

 Connecting Peptide / C-Peptide (Residues 55–89): Demonstrates high sequence divergence and insertion/deletion events (gaps), reflecting its primary role in protein folding rather than direct receptor binding.

 Insulin A-Chain (Residues 90–110): Shows near-absolute sequence preservation across all terrestrial mammals.

3.2 Graphical Alignment Visualization

The alignment overview confirms that the cysteine residues responsible for essential intra- and inter-chain disulfide bonds are strictly conserved across all 7 taxa.

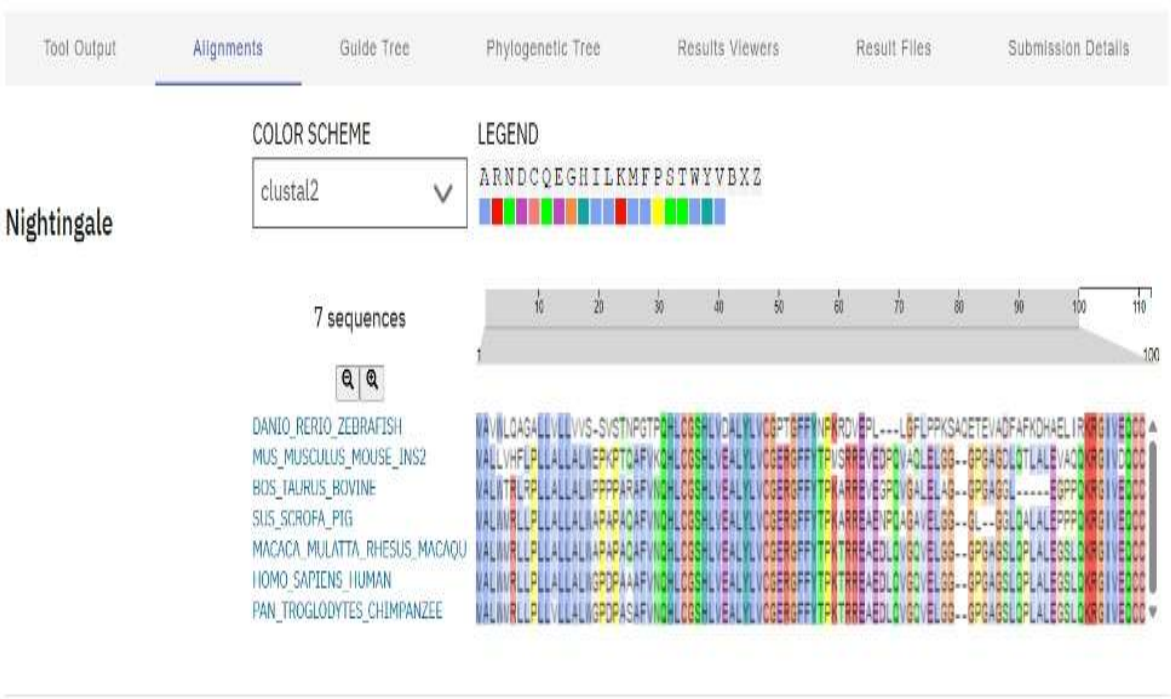


Figure 3: Interactive Alignment Graphical Overview (Nightingale Viewer):

4. Quantitative Similarity: Percent Identity Matrix

The pairwise identity matrix generated by Clustal2.1 quantifies the exact sequence similarity percentages across all pairs of analyzed species.

```
#
#
#  Percent Identity  Matrix - created by Clustal2.1
#
#

     1: Danio_rerio_Zebrafish          100.00   43.40   47.52   45.19   44.34   44.34   45.28
     2: Mus_musculus_Mouse_Ins2         43.40  100.00   78.10   78.70   81.82   80.91   80.00
     3: Bos_taurus_Bovine               47.52   78.10  100.00   86.41   83.81   83.81   82.86
     4: Sus_scrofa_Pig                  45.19   78.70   86.41  100.00   89.81   87.04   86.11
     5: Macaca_mulatta_Rhesus_macaque   44.34   81.82   83.81   89.81  100.00   97.27   96.36
     6: Homo_sapiens_Human              44.34   80.91   83.81   87.04   97.27  100.00   98.18
     7: Pan_troglodytes_Chimpanzee      45.28   80.00   82.86   86.11   96.36   98.18  100.00
```

Figure 4: Percent Identity Matrix for Preproinsulin Sequences:

4.1 Summary of Pairwise Identity Metrics

Species Pair Comparison	Sequence Identity (%)	Biological / Evolutionary Context
Human vs. Chimpanzee	98.18%	Extremely high orthology (2 amino acid differences)
Human vs. Rhesus Macaque	97.27%	High conservation within Simiiformes
Human vs. Pig	87.04%	Moderate mammalian conservation (Historical therapeutic use)
Human vs. Bovine	83.81%	Moderate mammalian conservation
Human vs. Mouse (Ins2)	80.91%	Conservation preserved in active chain domains
Human vs. Zebrafish	44.34%	Distant orthology (Teleost outgroup divergence)

5. Phylogenetic Tree & Evolutionary Reconstruction

The resulting phylogram accurately reflects established vertebrate evolutionary lineages:

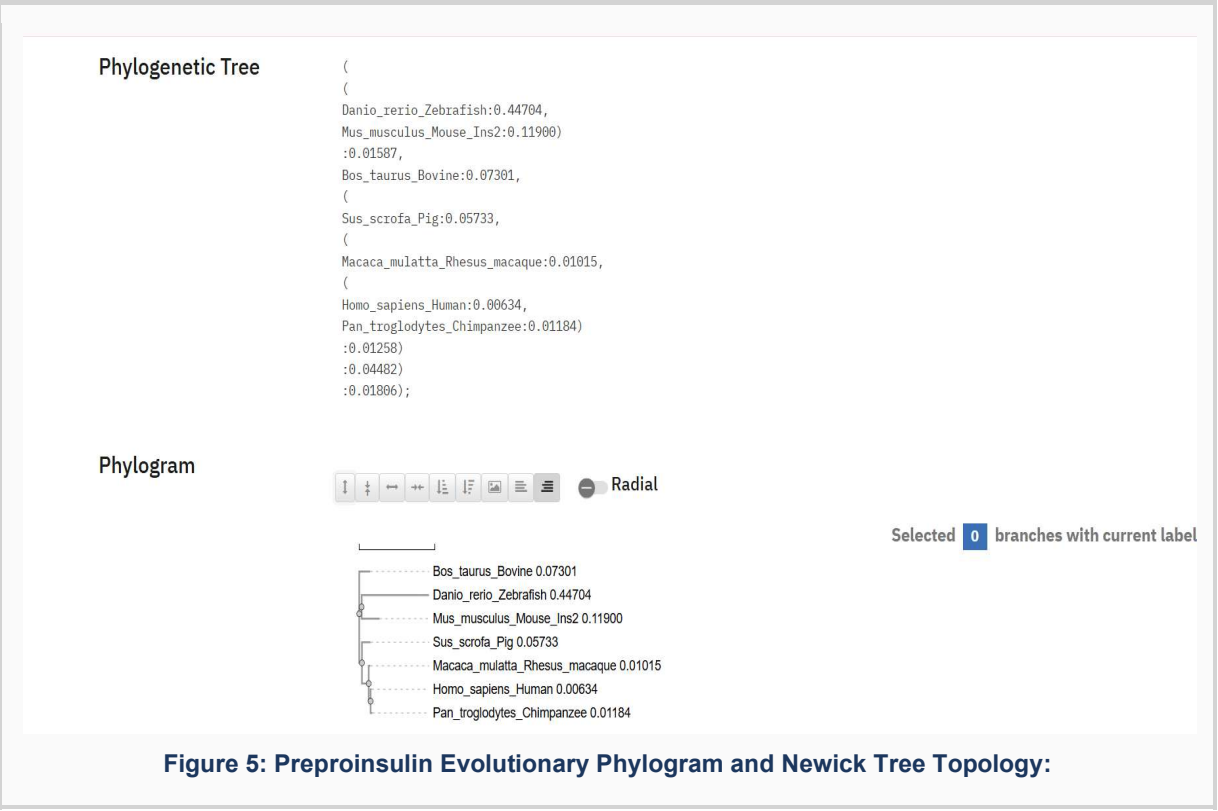


Figure 5: Preproinsulin Evolutionary Phylogram and Newick Tree Topology:

 Clade 1 (Primates): *Homo sapiens*, *Pan troglodytes*, and *Macaca mulatta* form a tightly clustered monophyletic clade with minimal branch lengths (0.00634 to 0.01184).

 Clade 2 (Ungulates): *Bos taurus* (0.07301) and *Sus scrofa* (0.05733) branch together, sharing an ancestor closer to primates than rodents.

 Clade 3 (Rodents): *Mus musculus* forms a distinct mammalian lineage.

 Outgroup: *Danio rerio* (0.44704) serves as the ancestral outgroup, branching at the base of the phylogram.

6. Conclusion

The Clustal Omega multiple sequence alignment and phylogenetic reconstruction of preproinsulin confirm strong purifying selective pressures on active hormone chains (A and B chains) across mammals. While the non-catalytic C-peptide undergoes evolutionary drift, critical catalytic and structural residues—most notably the six conserved cysteines—are preserved to maintain insulin receptor binding kinetics across hundreds of millions of years.

References / Data Sources:

1. EMBL-EBI Clustal Omega Tool: https://www.ebi.ac.uk/Tools/msa/clustalo/

2. Sievers, F., et al. Fast, scalable generation of high-quality protein multiple sequence alignments using Clustal Omega. Molecular Systems Biology, 2011;7(1):539.
3. UniProt Database for ortholog sequence retrieval: https://www.uniprot.org/